Universiteit van Amsterdam
Data Science Centre

INVITED TALK

Hyperbolic Deep Learning

What, Why and How



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HAVA Lab | University of Amsterdam

HELLO

A quick hello

India

Bachelors & Masters

IIIT Bangalore

Adobe Research

Research Associate

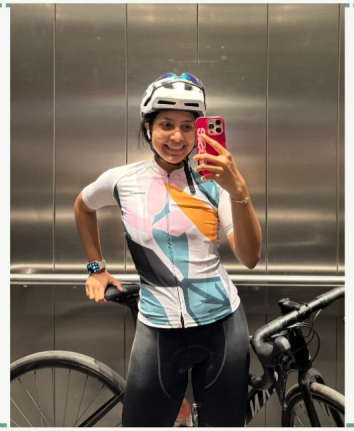
Image, video and
layout generation

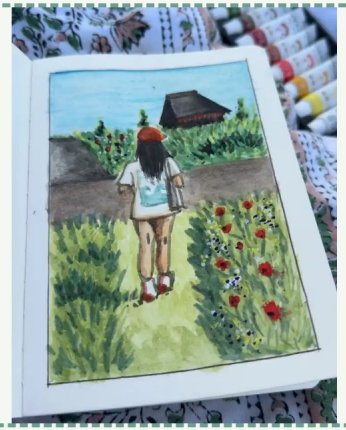
PhD

AI for dentistry

University of Amsterdam

Outside the lab







WHAT

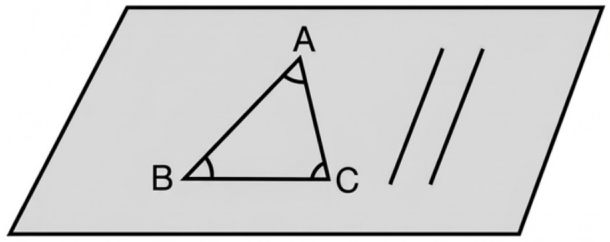
Who has heard of hyperbolic geometry?

Or Escher's artworks?



WHAT

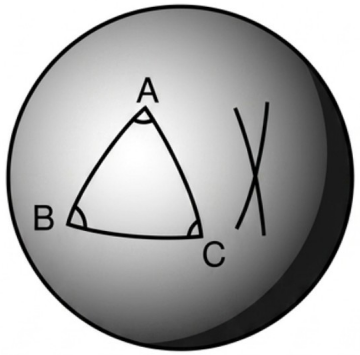
Three geometries



$\hat{A} + \hat{B} + \hat{C} = 180°$

Euclidean

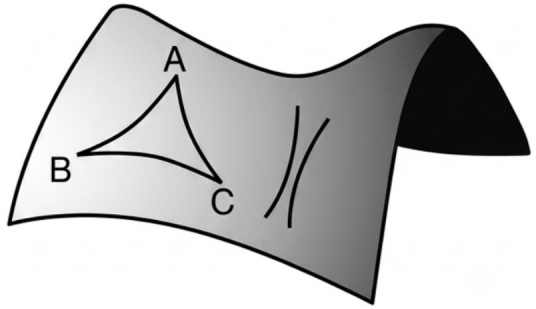
Zero curvature



$\hat{A} + \hat{B} + \hat{C} > 180°$

Spherical

Positive curvature



$\hat{A} + \hat{B} + \hat{C} < 180°$

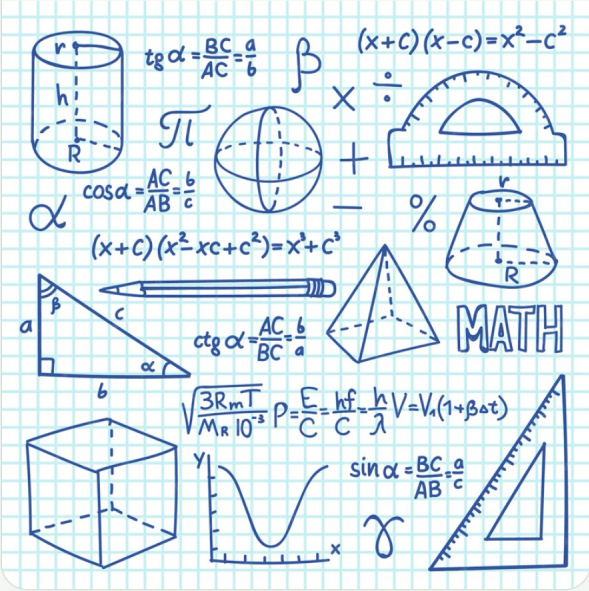
Hyperbolic

Negative curvature

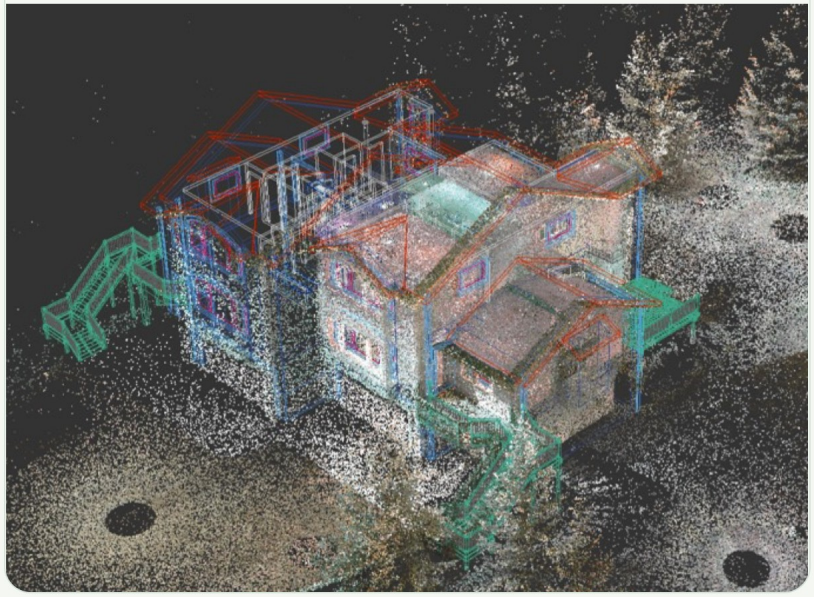
WHAT · THE DEFAULT

Is the Earth flat?

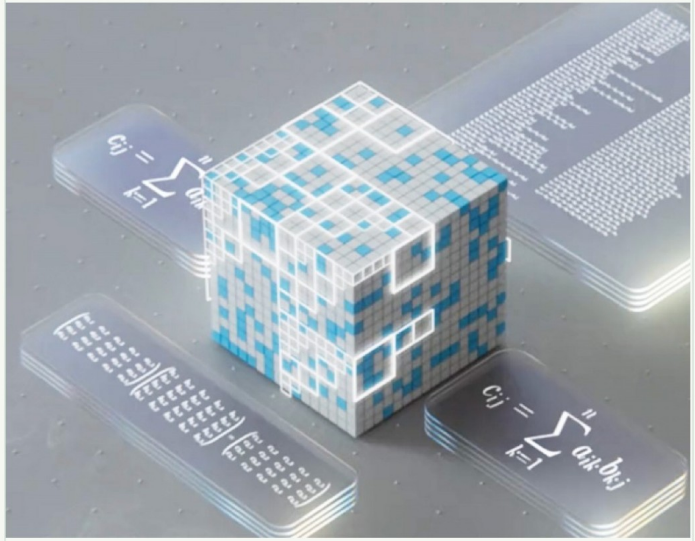
It's what we were taught



Data can be euclidean



It's what our GPUs expect



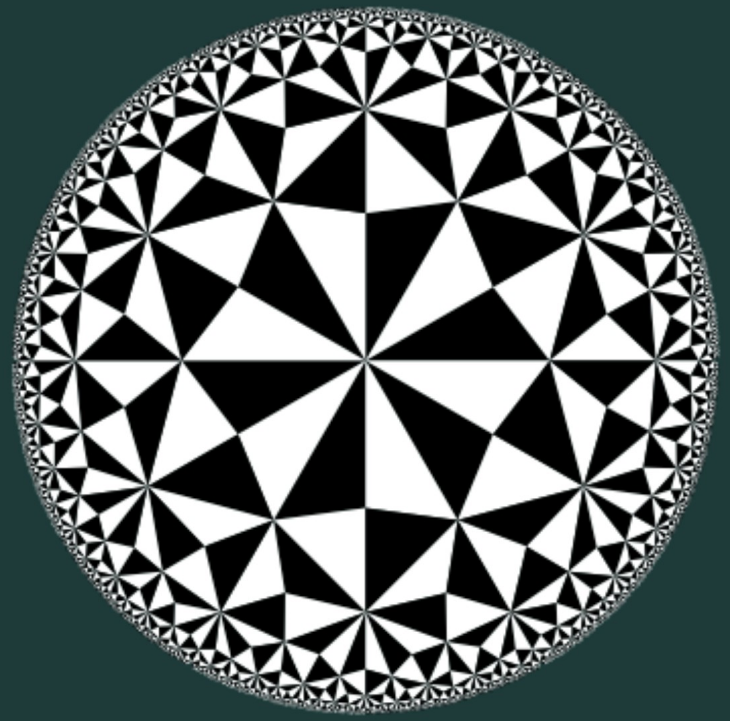
Fawzi et al. Nature 2022

Euclidean space is the implicit default of deep learning.

WHY

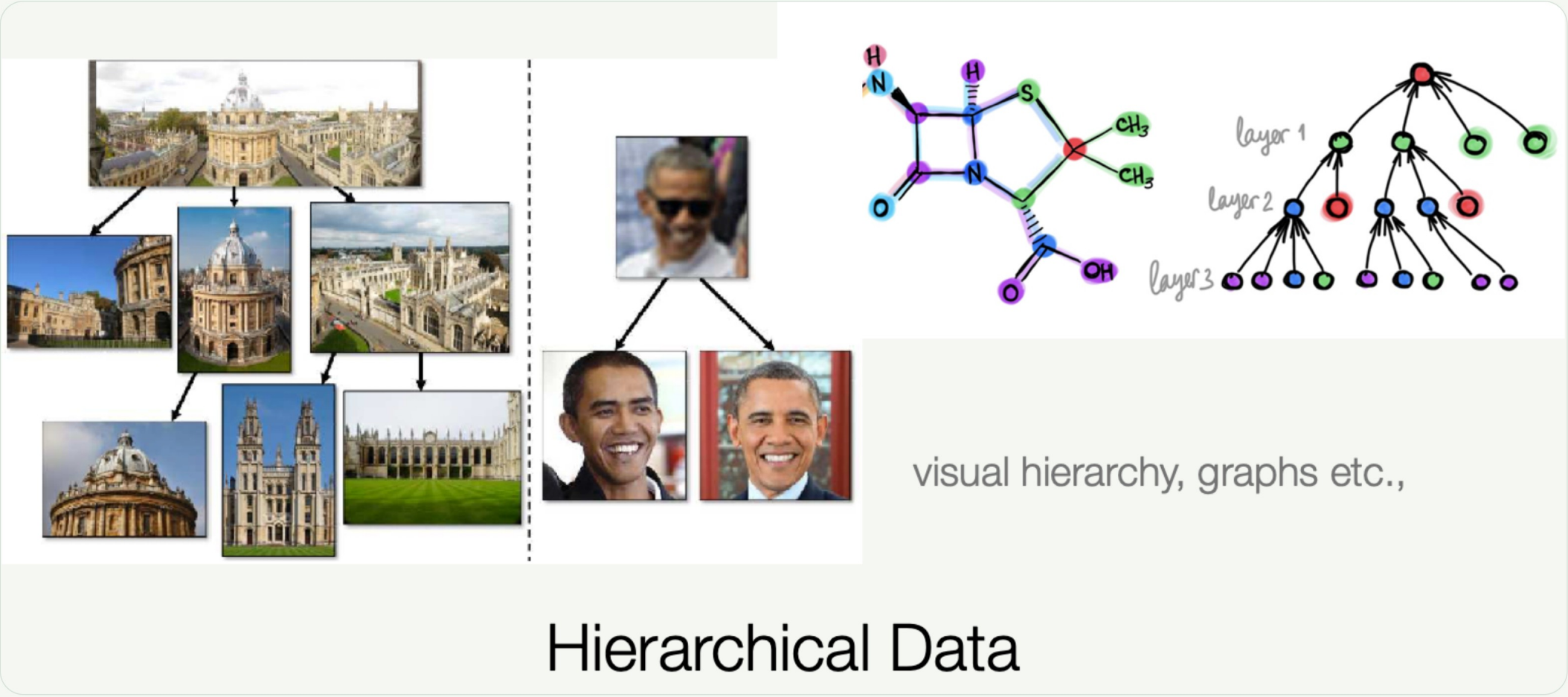
Is flat really the best choice?

When are straight lines not enough?



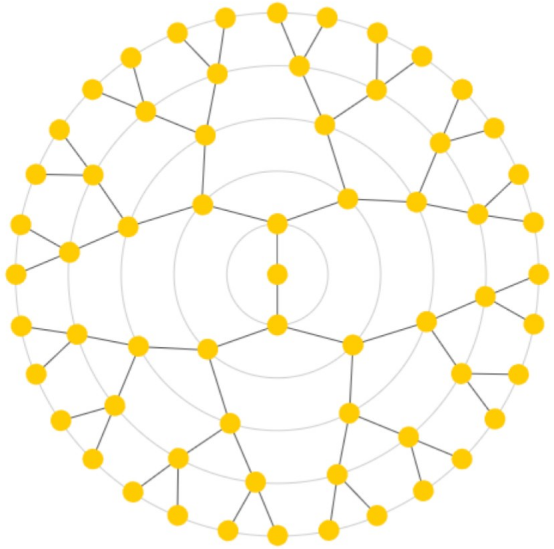
WHY · HIERARCHY

Structure in data

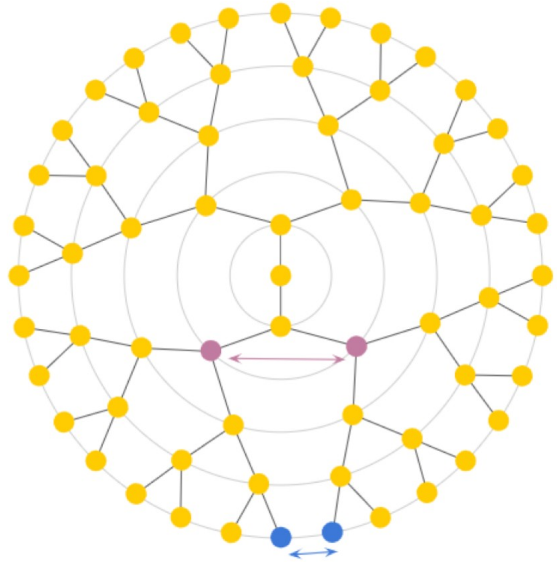


WHY · DISTORTION

Trees outgrow flat space



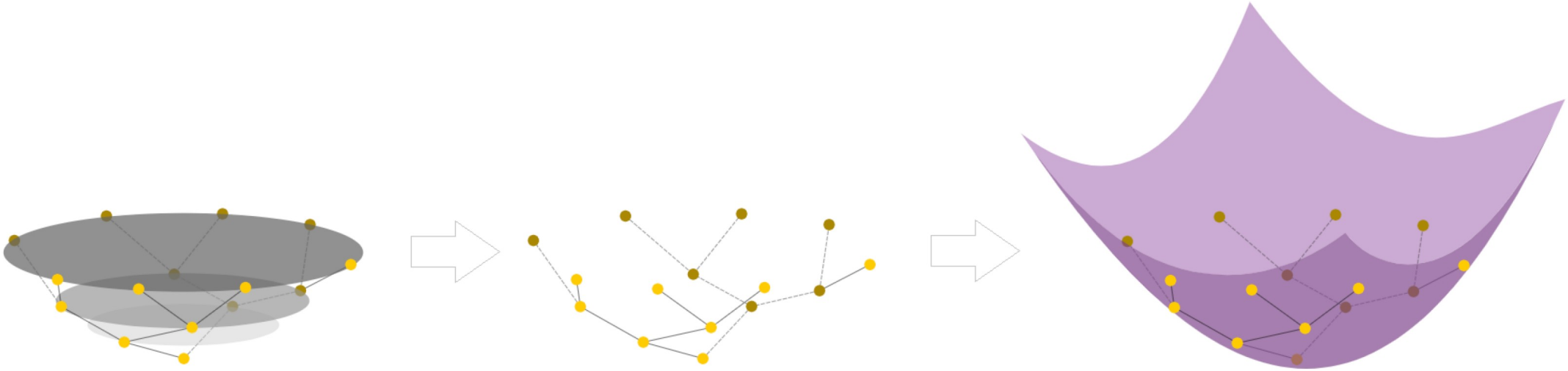
Every level adds exponentially more nodes



- Siblings pushed apart
- Distant leaves pushed together

WHY · HYPERBOLIC

Hyperbolic space grows like a tree



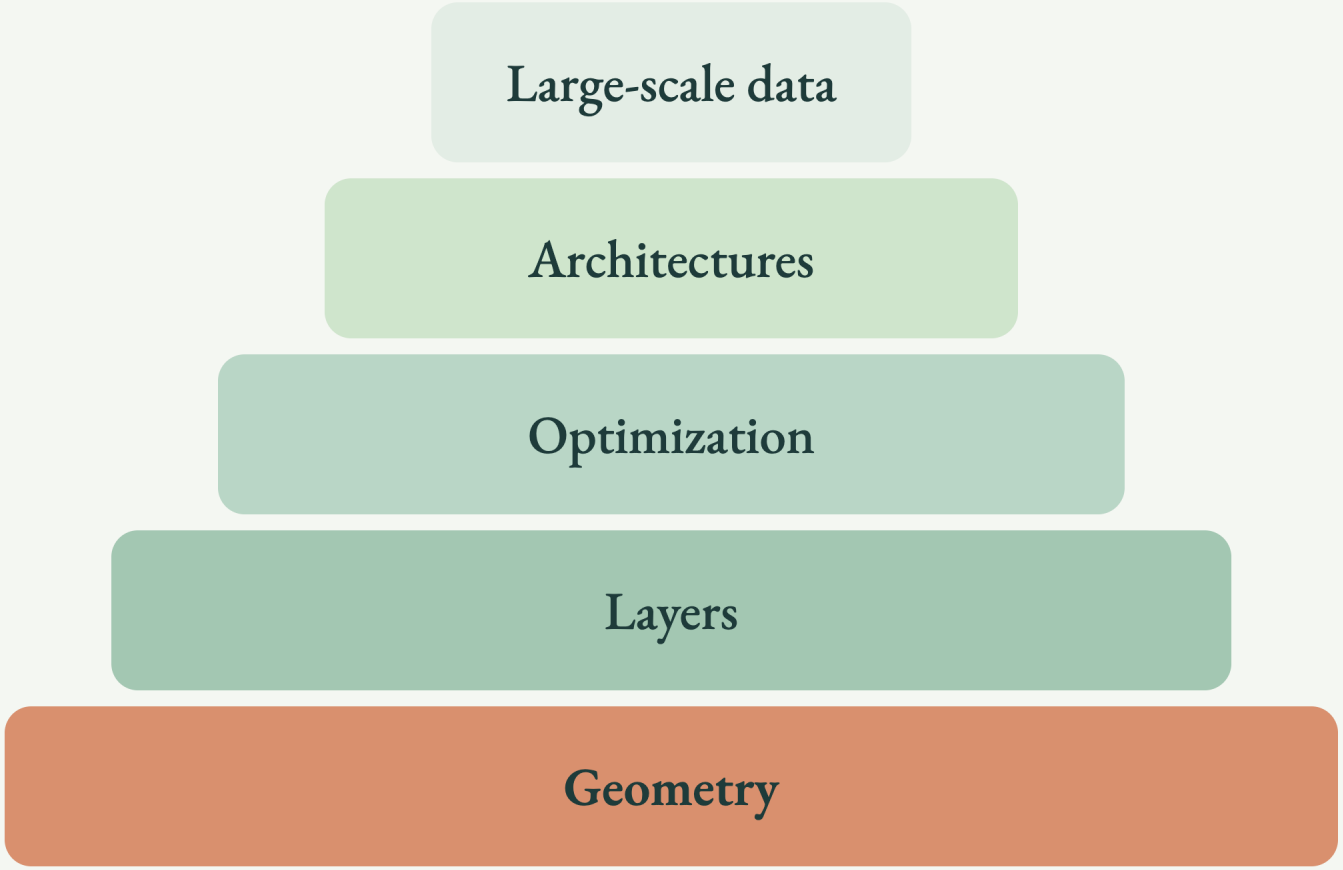
Negative curvature

Volume grows exponentially

Trees embed with low distortion

HOW

Hierarchy of deep learning



Change the bottom, and everything above has to adapt.

PAPER 1 · INTERPRETABILITY

Hyperbolic Concept Bottleneck Models

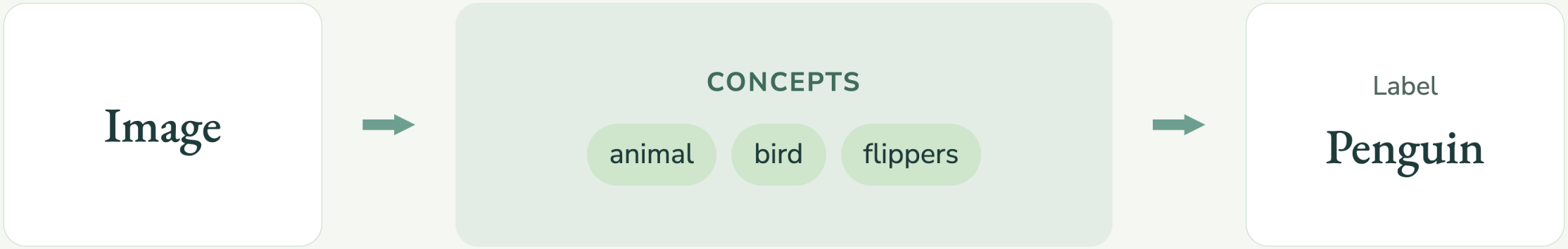
Daniel Uyterlinde, **Swasti Shreya Mishra**, Pascal Mettes



NeurIPS 2026 · Spotlight

HOW · PAPER 1

Concept bottleneck models



Predict through human-readable concepts

But concepts sit in flat space as independent axes

WHY · PAPER 1

Concepts are hierarchical

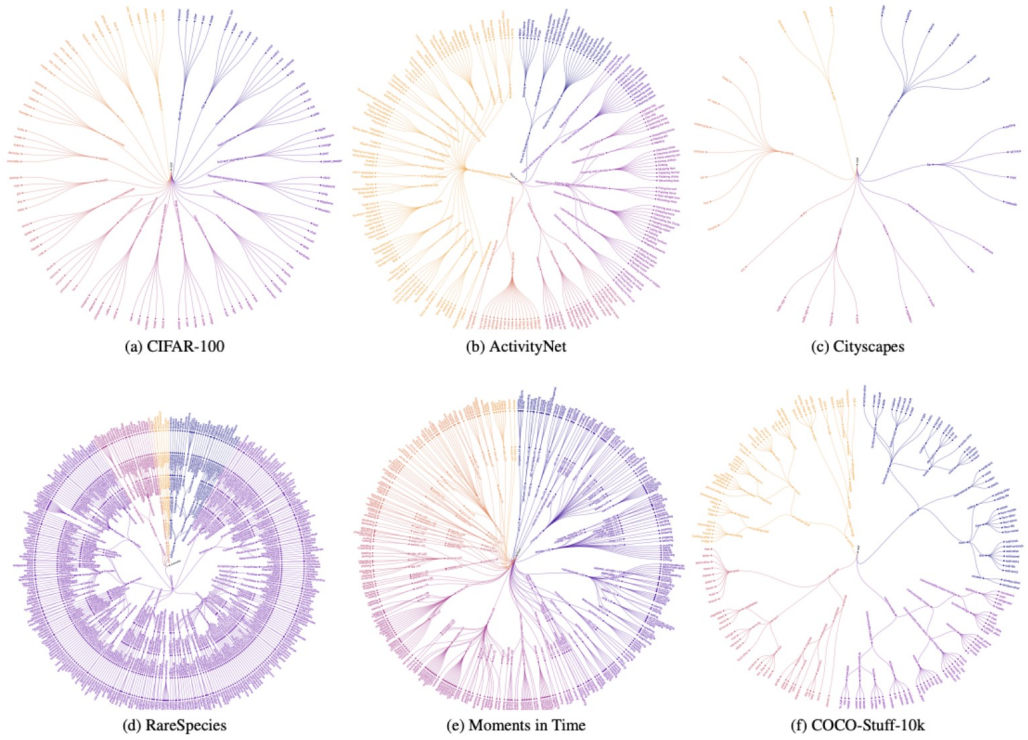


Figure 1. Hierarchy visualizations for a selection of datasets form *HierVision*. The coverage in the repository hub ranges from small number of classes and shallow trees to very large number of classes and deep trees.

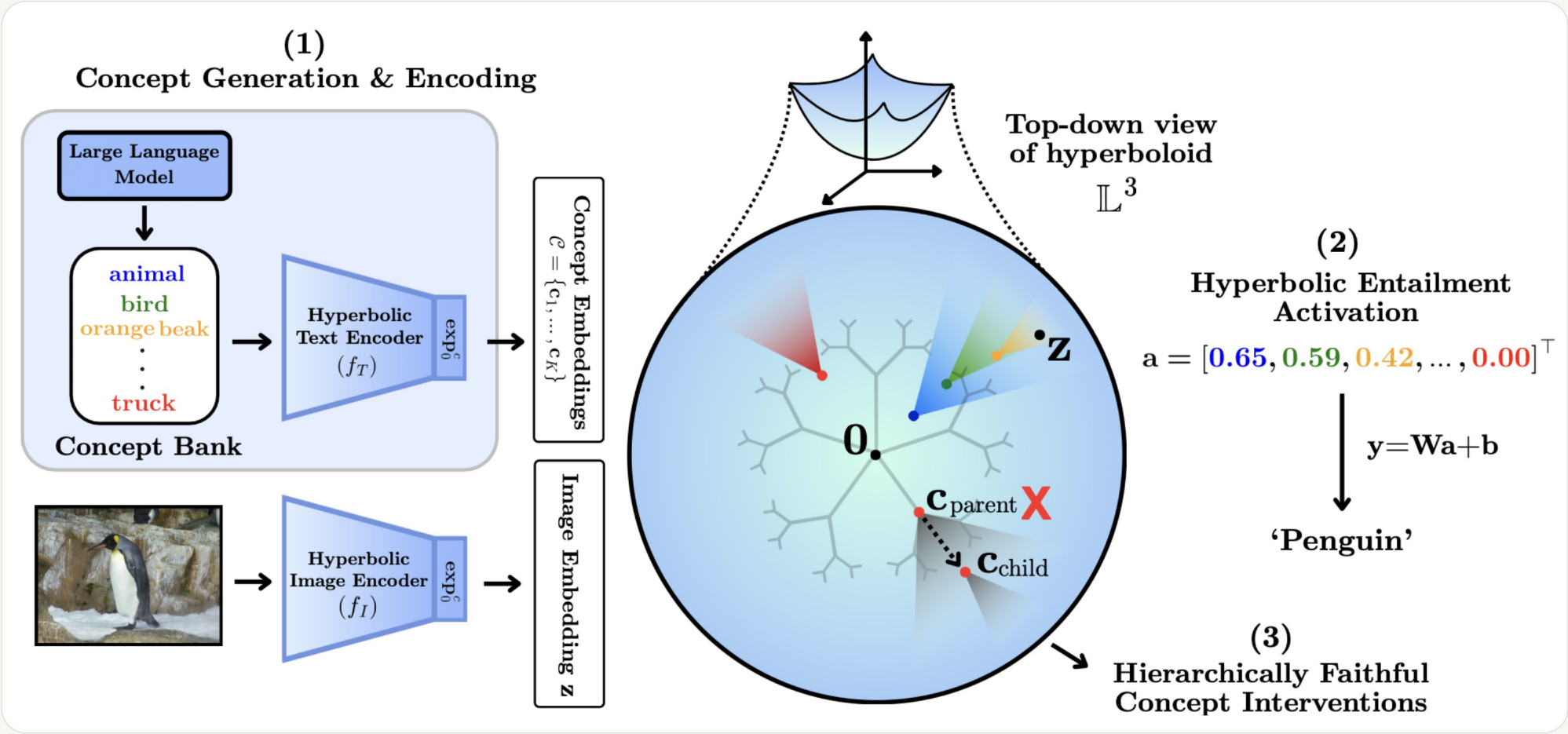
Taxonomies, not flat lists

Parents contain their children

Flat Euclidean concepts ignore this

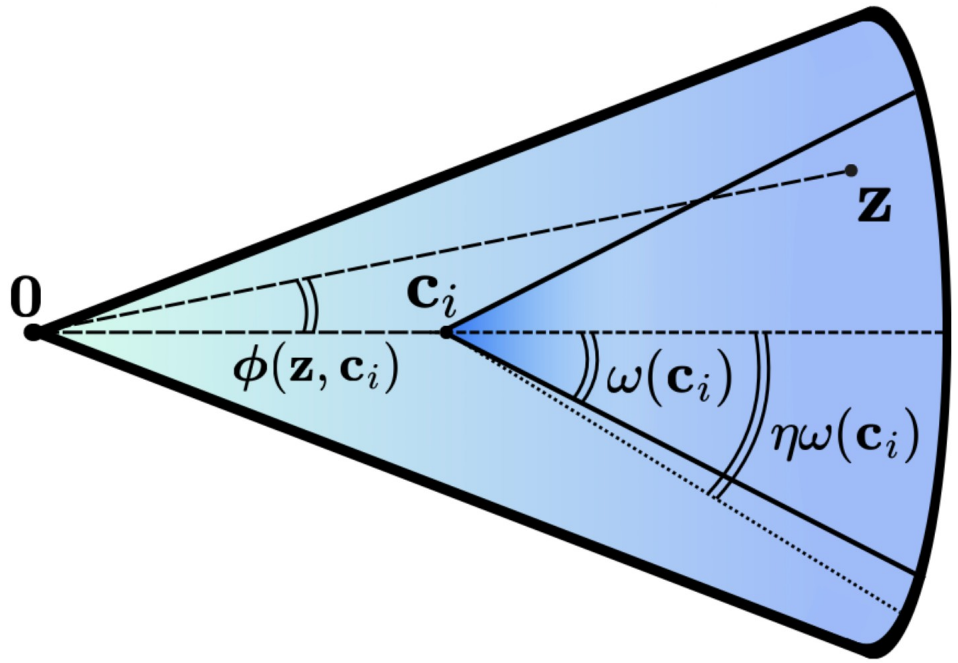
HOW · PAPER 1

HypCBM at a glance



HOW · PAPER 1

Concept activation for free



(a) Geometric Entailment

1. **Concepts are cones**
   A parent concept contains its children

2. **Inside means active**
   Is the image embedding inside the cone?

3. **Margin is the activation**
   Sparse and hierarchy-aware, with no extra training

HOW · PAPER 1

What we gain

20×

more data used by the Euclidean models it rivals

Coherent

hierarchy-aware activations and interventions that follow the tree

Robust

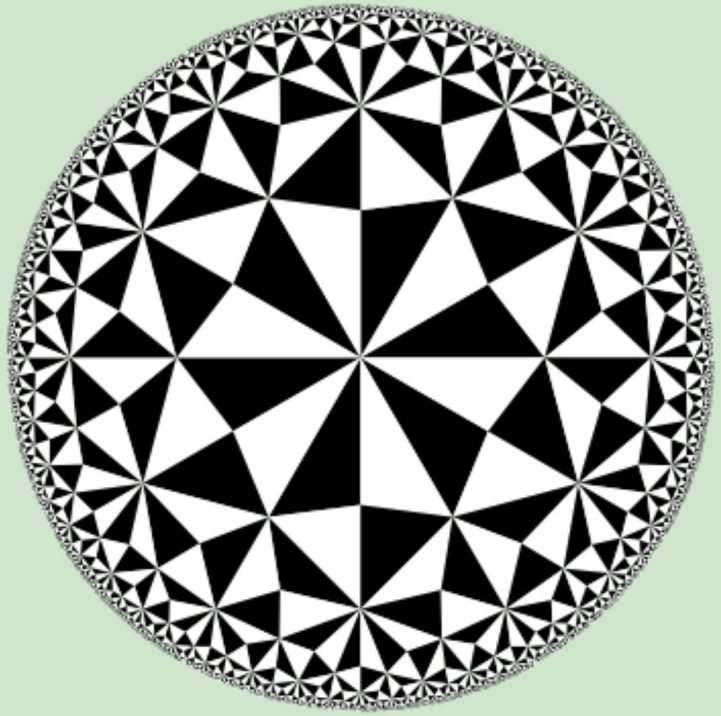
better under input corruptions

Dataset	Method	Backbone	Accuracy @ Concept Budget (ANEC-K)						
			5	10	20	50	100	Peak	LP
CIFAR-100 ($N_c = 613$)	LF-CBM	CLIP-20M	39.7	48.9	55.4	60.6	64.5	67.9	73.1
	LF-CBM	HyCoCLIP-20M	25.6	34.6	56.3	61.6	62.6	69.5	76.3
	HypCBM (Ours)	HyCoCLIP-20M	**44.3**	**54.4**	**60.9**	**69.7**	**71.8**	**72.2**	76.3
	Ref: LF-CBM	*CLIP-400M*	*28.0*	*58.0*	*63.9*	*64.7*	*67.0*	*71.3*	*82.6*
SUN397 ($N_c = 1935$)	LF-CBM	CLIP-20M	45.8	53.4	61.3	67.0	68.6	70.5	74.8
	LF-CBM	HyCoCLIP-20M	34.3	44.2	50.4	59.1	61.5	72.1	75.9
	HypCBM (Ours)	HyCoCLIP-20M	**49.3**	**58.3**	**65.7**	**69.9**	**71.3**	**72.3**	75.9
	Ref: LF-CBM	*CLIP-400M*	*40.0*	*48.0*	*58.8*	*64.4*	*66.8*	*79.0*	*82.7*
ImageNet ($N_c = 2950$)	LF-CBM	CLIP-20M	7.7	15.3	30.7	47.3	53.7	55.3	63.7
	LF-CBM	HyCoCLIP-20M	10.2	15.5	26.0	40.6	46.2	57.9	65.4
	HypCBM (Ours)	HyCoCLIP-20M	**17.9**	**34.6**	**44.2**	**52.3**	**56.8**	**58.1**	65.4
	Ref: LF-CBM	*CLIP-400M*	*17.4*	*34.0*	*51.8*	*52.6*	*54.0*	*73.8*	*78.5*

BEYOND HIERARCHIES

Is hierarchy the only reason to go hyperbolic?

Can geometry make models robust?



PAPER 2 · ROBUSTNESS

Hyperbolic U-Net for Robust Medical Image Segmentation

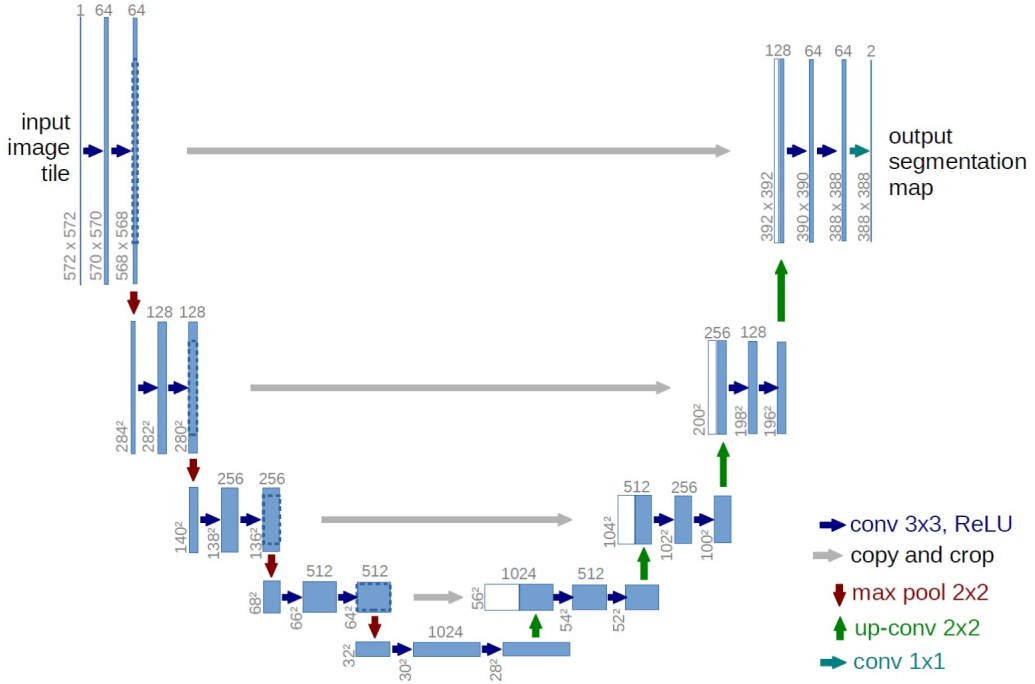
Swasti Shreya Mishra, Max van Spengler, Erwin Berkhout, Pascal Mettes



MIDL 2026

WHY · PAPER 2

U-Net still reigns



U-Net: Convolutional Networks for Biomedical Image Segmentation

447k Accesses 105k Citations 99 Altmetric

nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation

Fabian Isensee, Paul F. Jaeger, Simon A. A. Kohl, Jens Petersen & Klaus H. Maier-Hein

Nature Methods **18**, 203–211 (2021) | Cite this article

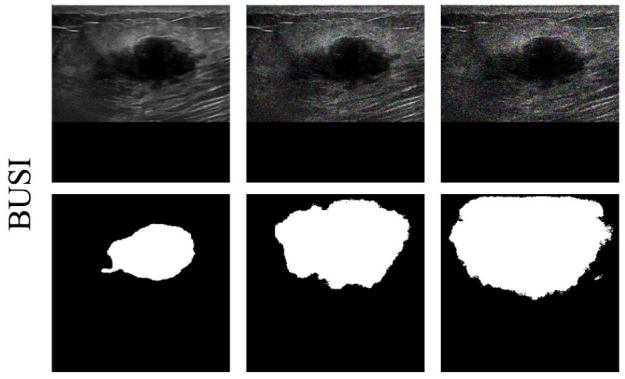
Save article

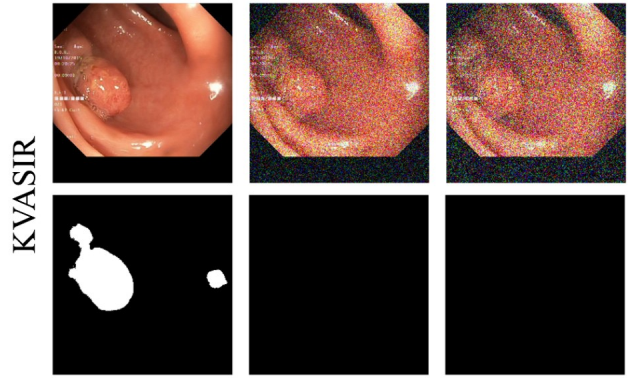
191k Accesses | **10k** Citations | **230** Altmetric | Metrics

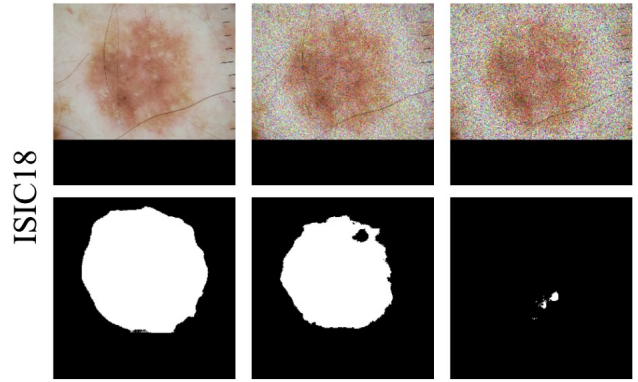
Still the backbone of medical image segmentation.

WHY · PAPER 2

The hidden flaw



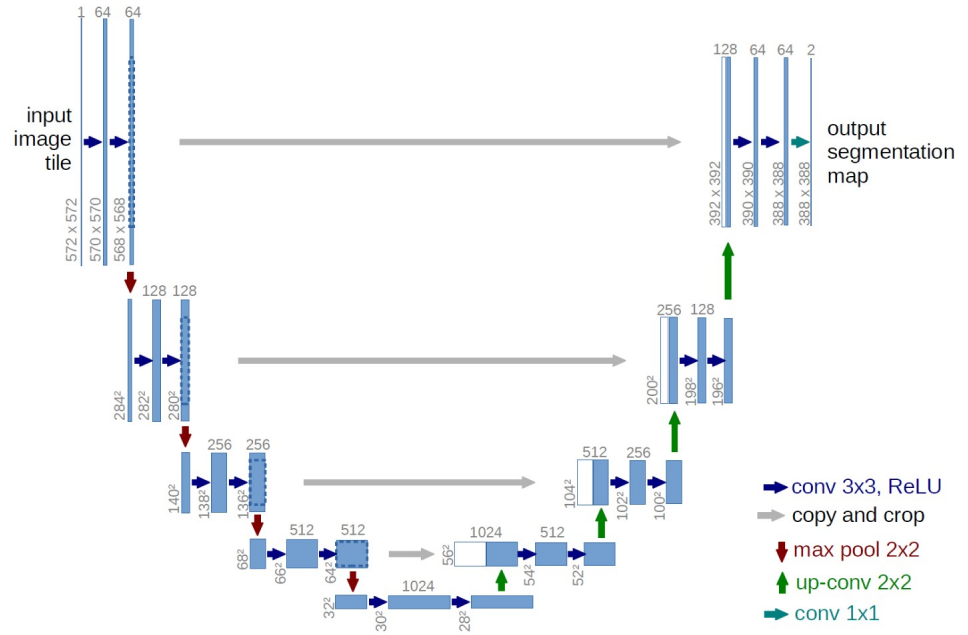




Segmentations collapse as noise grows.

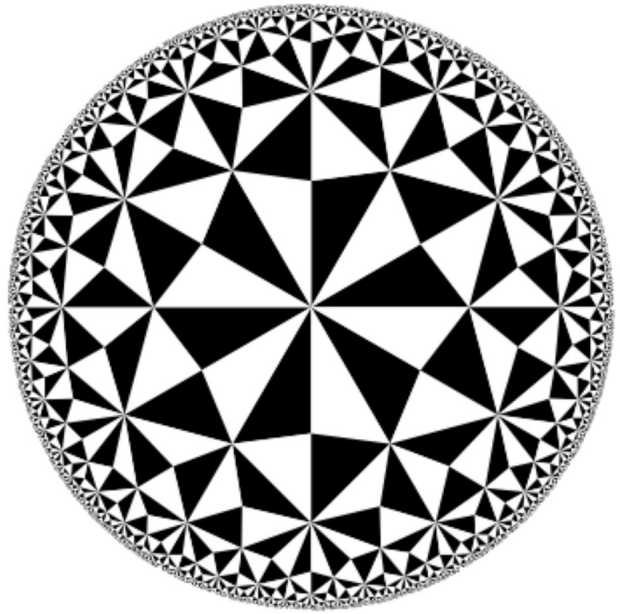
WHY · PAPER 2

What if the problem isn't the network?



U-Net

+



Hyperbolic Poincaré disk

HOW · PAPER 2

Two roadblocks, two fixes

ROADBLOCK 1

No hyperbolic decoders

Fully hyperbolic work has focused on encoders

OUR FIX

Hyperbolic transposed convolution

Hyperbolic bilinear upsampling

ROADBLOCK 2

Unsuitable initialization

Existing hyperbolic initializations don’t fit decoder blocks

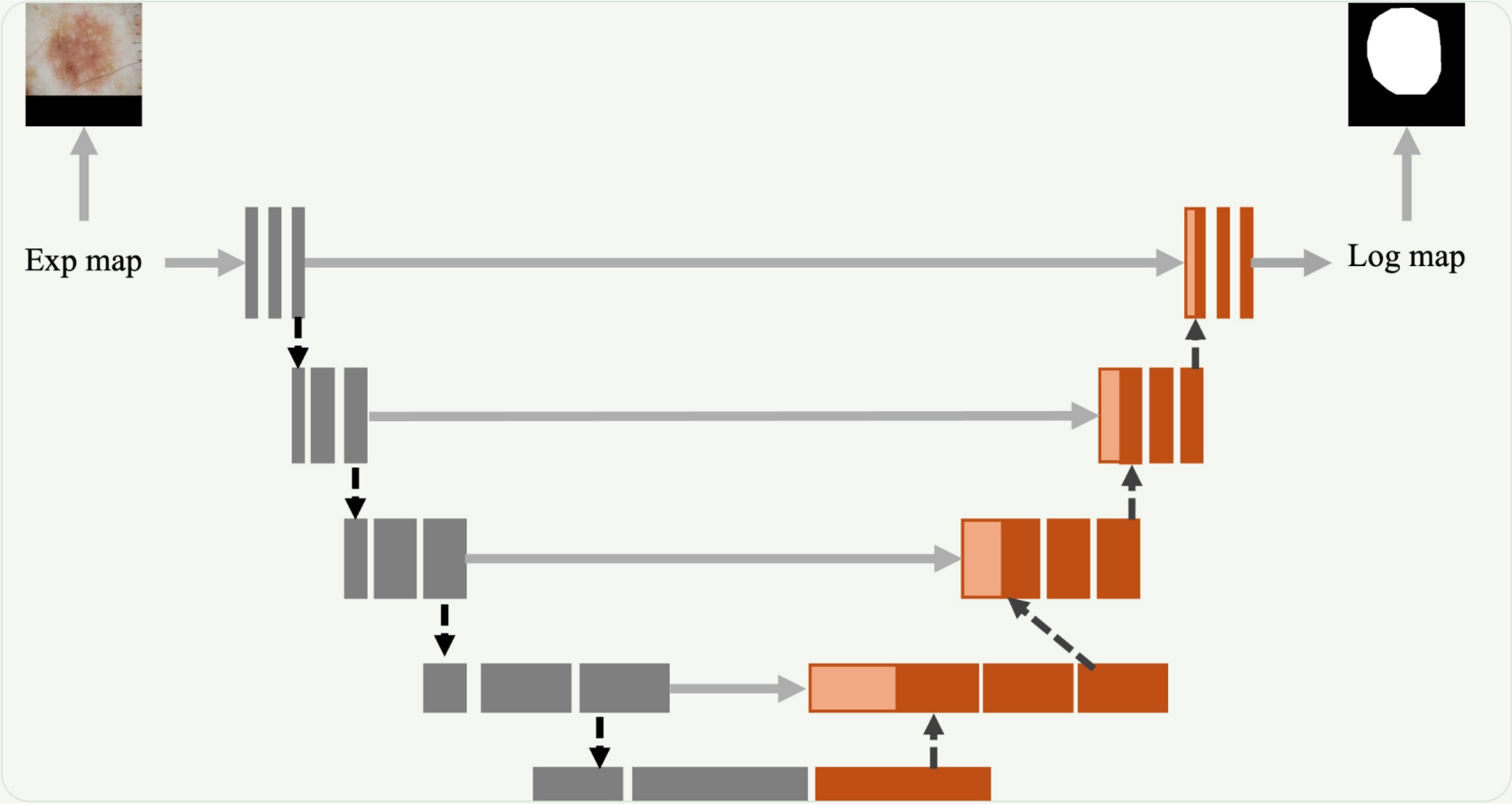
OUR FIX

Newton-approximation scaled initialization

Norms preserved at the start of training

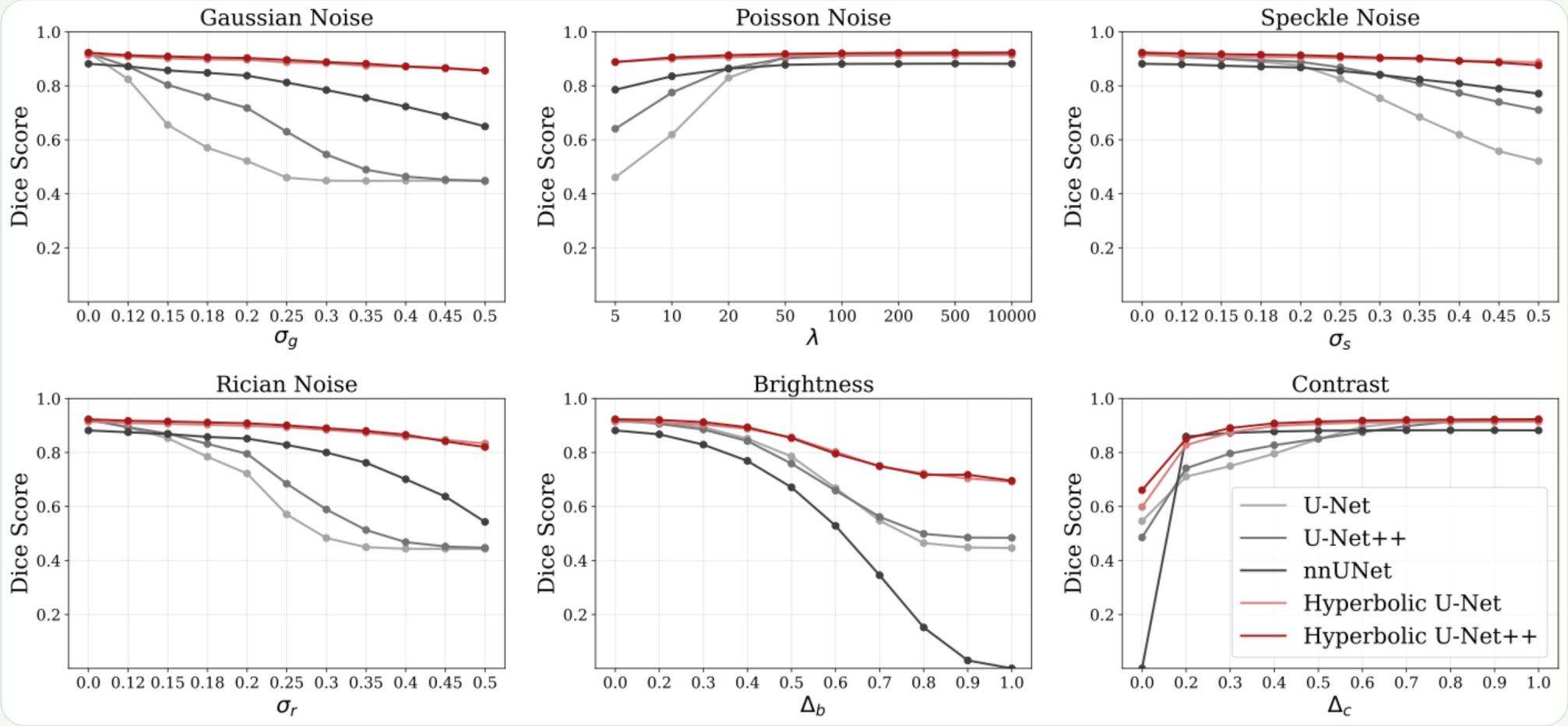
HOW · PAPER 2

The first fully hyperbolic U-Net



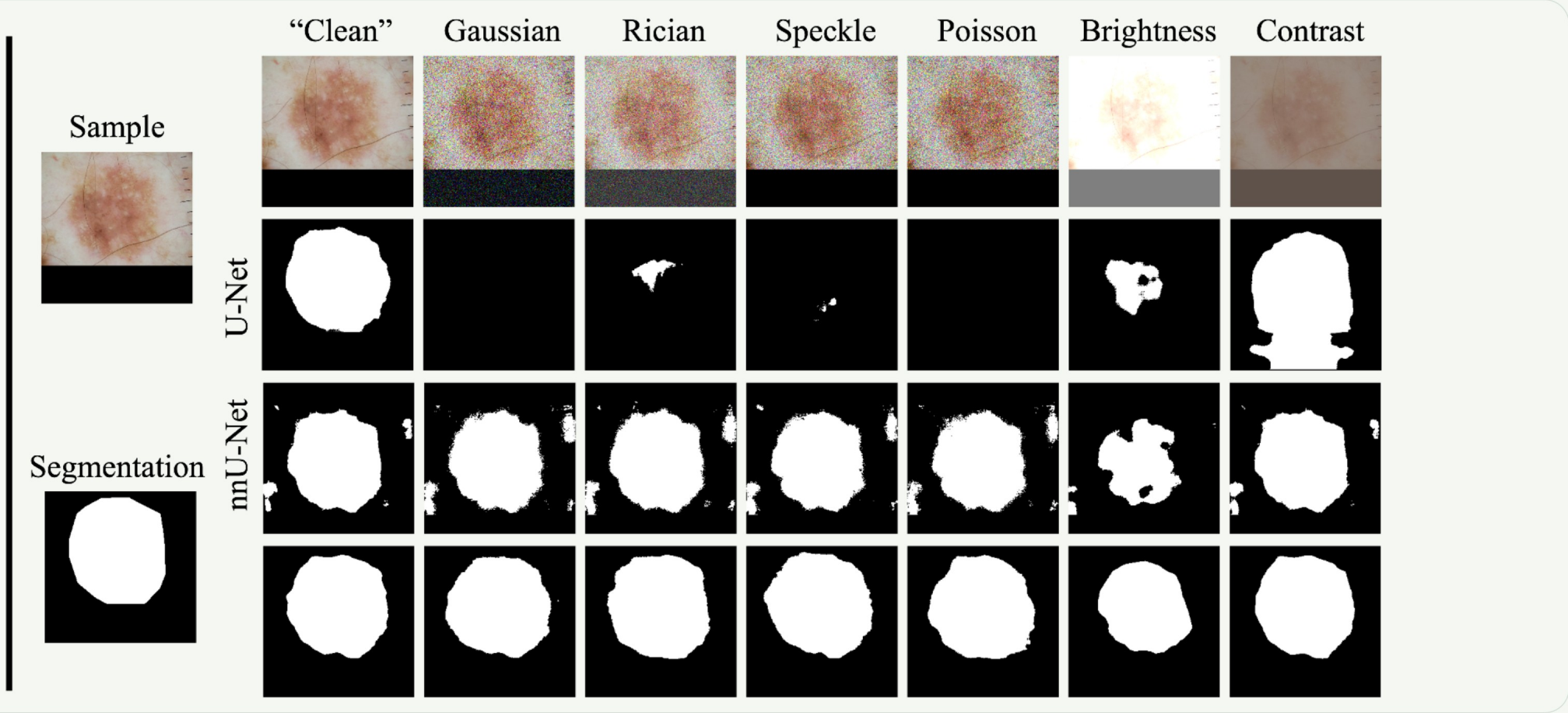
RESULT · PAPER 2

Hyperbolic U-Net stays robust



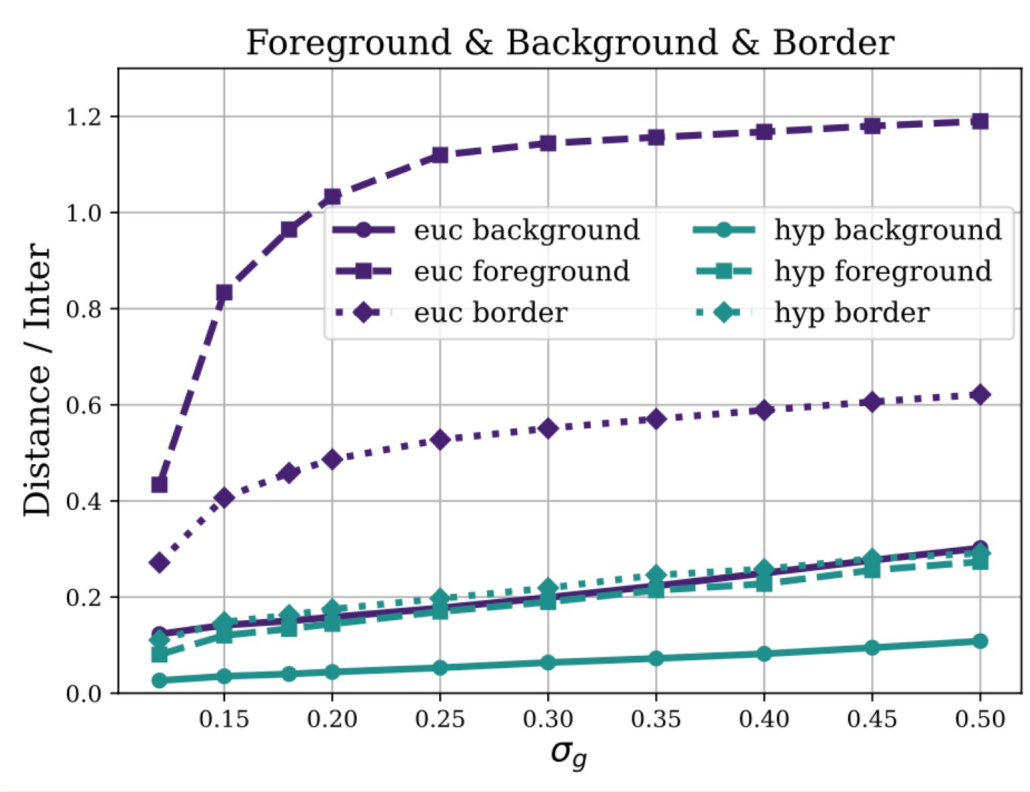
RESULT · PAPER 2

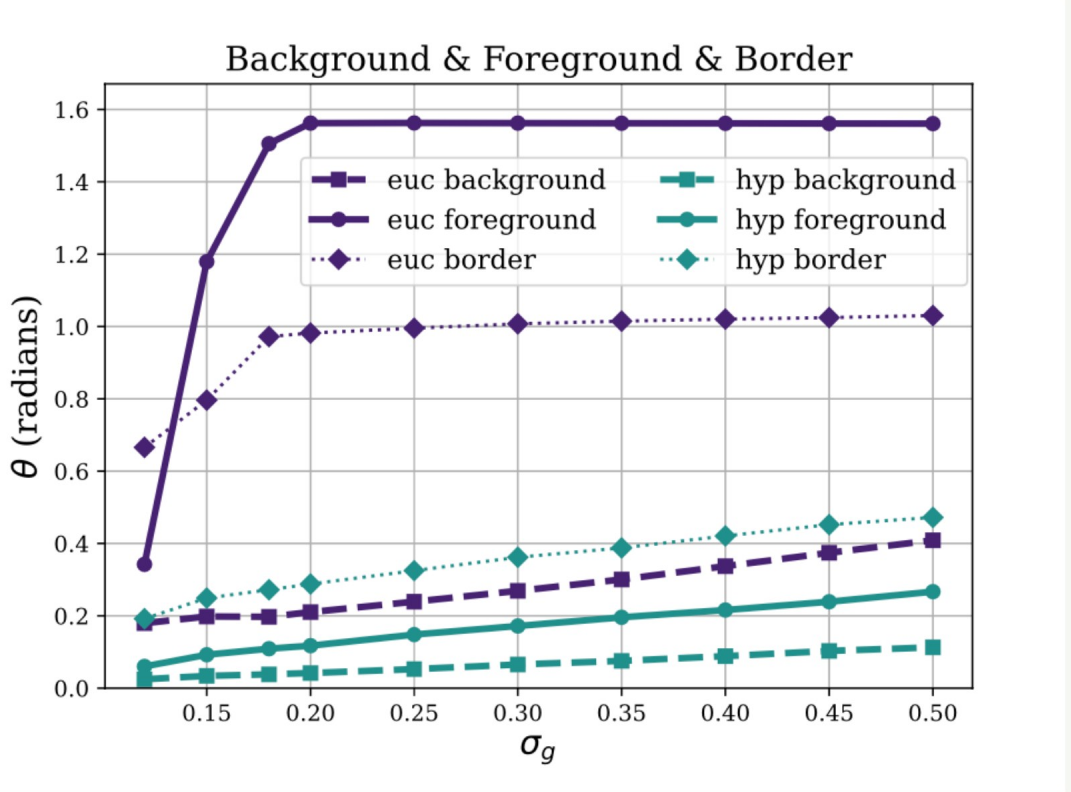
Segmentations hold under noise



WHY · PAPER 2

Why does hyperbolic geometry help?





TAKEAWAYS

The Earth isn’t flat, and neither is data

WHAT

Geometry is a design choice

Euclidean is a default, not a given

WHY

Structure and robustness

Hierarchy gives interpretability, and beyond it, robustness

HOW

Light or all-in

Add geometry post-hoc, or rebuild every layer

Open directions: scaling up · hyperbolic foundation models · clinical settings

Thank you!

Questions?

GitHub: swastishreya

Email: s.s.mishra@uva.nl

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Data Science Centre

